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What is the effect of Lambda-cyhalothrin concentration (mM) on the moles of oxygen produced after 5 minutes of peroxide decomposition catalyzed by *Raphanus sativus* peroxidase measured by the transmittance (%) of the solution when mixed with guaiacol?

Background:
Enzymes, or proteins which catalyze metabolism, are vital for the survival of all living organisms. Without them, chemical reactions within living beings would take years to occur. Like all catalysts, enzymes lower the activation energy of specific chemical reaction without themselves being consumed or altered by the reaction. During reactions, enzymes act upon molecules called substrates to form products. Interactions between enzyme active sites and substrates occur at the active sites which are clefts or grooves on enzyme surfaces whose shapes and chemical properties are dependent on the enzyme tertiary structures. Once a substrate binds to an active site, the enzymes alter substrate conformation which in turns accelerate its transformation into the desired product. Due to the highly specific nature of the interaction, even slight changes to active sites will halt enzyme activities(Cooper 2020).

Raphanus sativus, more commonly known as radish, is widely available throughout the world. The root vegetable contains peroxidase (POD), a type of enzyme responsible for catalyzing the decomposition of the toxic, metabolic, waste product peroxide (H_2O_2) into harmless molecules (Gulcin, *et al.*). During the process, two moles of hydrogen peroxide are transformed into two moles of water and one mole of water (Everse 2014):

$$2\ H_2O_2 \rightarrow 2H_2O + O_2$$

A common radish POD inhibitor is cyanide, or any chemical compound which includes one or more CN groups (Chen *et al*). Inhibitors are molecules which prevent enzymes from working in a normal manner by either blocking or altering the shape of an enzyme's active site (Kuddus 2019). Research by Chen *et al.* shows that cyanide works by replacing the water in the six octahedral position of POD's iron heme group (2000). This is considered the active site of the POD (LibreTexts 2020). By altering the chemical structure of POD active sites, cyanide inhibits POD activities.

Guaiacol is an indicator of POD's oxygen formation. It reacts with oxygen to form guaiacol dehydrogenation product (GDHP) which has a brown colour. A darker POD-H_2O_2 solution with guaiacol added would indicate a higher level of oxygen produced which in turns indicates a higher level of enzyme activities. The opposite is true when the solution is lighter. The level of GDHP formed can be measured spectrophotometrically (Pütter 1974). The more GDHP, the lower the transmittance.

Pesticides protect plants from pests and increase crop production. However, when misused, these chemicals can have deadly impacts on their surrounding environments. A 2011 research on common pesticides' impact on pH regulatory enzymes in sea bass shows that pesticides are effective inhibitors for the enzymes (Aksakal *et al.* 2009). Pesticides can be also be toxic for plant POD. One of the most common example of pesticide is Lambda-cyhalothrin. It belongs to a class of pyrethroid chemicals, which has a toxic effect on the nervous systems of insects. It also contains a cyanide group. Research on the effects of Lambda-cyhalothrin on turnip POD demonstrates that, even at low concentrations, the chemical affects the shape of POD active sites resulting in a strong inhibitory effect (Gulcin *et al.* 2018).

Hypothesis:
As lambda-cyhalothrin concentration increases, the transmittance of the peroxide-peroxidase solution also increases. This is because lambda-cyhalothrin is a POD inhibitor which prevents peroxide from binding to the enzyme's active site. This disrupts the catalyzing of decomposition of H_2O_2 into H_2O and O_2, slowing

down oxygen production rate. Thus, there will be less GDHP and a higher transmittance rate after 5 minutes of reaction.

Methodology:
Preliminary experiments were conducted a week before the lab and they tested the optimal conditions for radish POD conditions. The optimal pH discovered is 6. There should also be 5ml of 4% POD per 0.3. ml 0.1% hydrogen peroxide for optimal POD activities.

The independent variable is Lambda-cyhalothrin concentration measured in mM (at a value of 5, 10, 15, 20, and 25). The values for the IV lie in the same range as the Gulcin *et al.*'s study on the effect of Lambda-cyhalothrin on turnip POD: 5mM to 25mM. This is done to see if the same moles of Lambda-cyhalothrin would have similar effects on a different type of root vegetable POD (radish). The result of this experiment can be use as a way to gain a better understanding on how similar different root vegetable PODs are. It also allows one to see how dangerous Lambda-cyhalothrin can be at relatively low concentrations for root vegetables. The dependent variable is transmittance rate of solution after 5 minutes of reaction. This is measured by a spectrometer.

Control variable	Why should it be controlled?	How to control it?
Temperature	A rise in temperature results in faster reaction rate as it increases kinetic energy of POD and H_2O_2 resulting in more collision and binding between POD active site and H_2O_2. Also, if temperature is too high it would disrupt the IMF within the POD, causing the POD to denature and transmittance to increase.	Conduct the experiment in a room which does not experience changes in temperature. Also keep all solutions used in the experiment at room temperature.
pH	Hydrogen bondings in POD's tertiary structures are sensitive to changes in pH. If not at optimal, there will be changes in POD's active sites. Thus, POD activities will slow and transmittance will increase.	As Lambda-cyhalothrin is acidic (pH 5), a pH 7 buffer should be added to the POD-H_2O_2 mixtures (5.5ml per trial). This allows the mixtures to remain at POD's optimal pH of 6.
Salinity	Ionic bonding in the POD's tertiary structure is sensitive to changes in salinity. Thus, changes in salinity can affect POD active site and slow down POD activities, increasing transmittance.	No salt should be added to the POD-H_2O_2 mixtures.
POD concentration	More POD means that H_2O_2 are more likely to collide with a POD active site, leading to more POD activities and lower transmittance.	The volume of POD should be the same for all trials (2 ml). Volume for all other solutions (guaiacol, Lambda-cyhalothrin and H_2O_2) should also be kept constant.
H2O2 concentration	More H_2O_2 means a higher chance for collision between H_2O_2 and POD active sites, leading to more POD activities and lower transmittance.	The volume of H_2O_2 should be the same for all trails (0.3ml). Also concentration of the solution should be 0.1%.

Control variable (continues)	Why should it be controlled?	How to control it?
Time POD is exposed to Lambda-cyhalothrin	Lambda-cyhalothrin is the IV of this experiment. The longer it is exposed to the POD, it more effect it would have on the transmittance.	Place POD and Lambda-cyhalothrin in separate test tubes during the preparation process. Use a timer make sure that POD and Lambda-cyhalothrin are only exposed to each other for 5 minutes before the mixture's transmittance is recorded.
Time POD is exposed to H_2O_2	The longer POD is exposed to H_2O_2 the more collisions occur between the substrate and the active site, leading to more POD activities and lowers transmittance.	Place POD and H_2O_2 in separate test tube during the preparation process. Use a timer to make sure that POD and H_2O_2 are only exposed to each other for 5 minutes before the mixture's transmittance is recorded.
Setting of vernier spectrometer	Inconsistency in spectrometer setting would lead to different recordings for transmittance.	Calibrate the machine with a cuvette filled with distilled water before the experiment. Keep the wave length the at 470 nm.

Procedure for extraction of 4% POD

Peel a radish then it cut into roughly 1 by 1 cm cubes with a knife. Weigh out 25g of the radish with the balance. Measure out 600 ml of distilled water, and pour into blender with the radish. Pulse the radish in 1−3 minute intervals for three times, with a 2 minutes pause between each time in the blender. Pour the mixture into a beaker over a filter cloth, and squeeze out all the remaining liquid in the cloth. Store the H_2O_2 extract in a refrigerator at 4 °C until time for lab.

Procedure for lab

*For all liquids measured with a graduated cylinder, look at the top of the liquid from eye level and take the measurement at the lowest point of the meniscus

*For all liquids measured with the syringe, keep the entire head of the syringe in the xdsolution to prevent air bubbles. If there are air bubbles, gently tap the syringe to move them to the top. Then push the the plunger to get rid of the bubbles (Vorvick 2020).

*Foaming occurs when Lambda-cyhalothrin is transferred from its container to the graduated cylinder, only measure the volume of liquid Lambda-cyhalothrin, then get rid of the foams with a pipette.

1. Turn on vernier spectrophotometer to let it warm up for 5 to 10 minutes
2. Calibrate the machine at the setting of 470 nm with a cuvettes filled to 2/3 with distilled water
3. Prepare baseline test by measuring out 2 ml of peroxidase and 5.5 ml of pH 7 buffer solution test tube and fill another test tube with 6.9 ml distilled water, 0.3 ml guaiacol, and 0.3 ml hydrogen peroxide. Measure each solution with the graduated cylinder and syringe (used to measure volume of guaiacol and hydrogen peroxide). Wash the apparatus before using them for a new solution. Made sure the test tubes are empty and clean before pouring in the solutions.
4. Combine the two test tubes then start a timer for 5 minutes
5. Place the mixture into a clean cuvette, fill it to about 2/3 full
6. Place the cuvette in to the spectrophotometer and record its transmittance once the timer gets to zero
7. Repeat step 3 to 6 four more times, use a different cuvette each time.

8. Repeat step 3 to 7 replace the water with 6.9 ml of 5 mM Lambda-cyhalothrin.
9. Repeat step 9 four more time, increase the molarity of the Lambda-cyhalothrin by 5 mM each time

Safety, ethical, and environmental issues:

Safety issues:

H_2O_2 can cause severe damage to one's eyes. Therefore eye protection should be worn at all time during the experiment. If the chemical made contact with eyes, flush them immediately with plenty of flowing water for 10 to 15 minutes with the eyelids held apart. If wearing contact lenses, remove them before flushing. If spilled, soak up with inert absorbent material (CarlRoth 2015).

Lambda-cyhalothrin can be poisonous when inhaled so the experiment should either be conducted in areas with constant air flow or with a mask on at all times. If inhaled, one should go to an area with fresh air. If breathing feels difficult, oxygen should be given. The chemical can also cause skin and eye irritation so chemical resistant gloves and protective eye protection should be worn during the experiment. If exposed, rinse the area under free flowing water for 10 to 15 minutes. If spilled on to clothes, take off the contaminated clothing and wash it before reuse. Keep in an enclosed container when not used to prevent spillage. If spilled, soak up with inert absorbent material (CarlRoth 2015).

If either H_2O_2 , Lambda-cyhalothrin, or guaiacol is swallowed during the experiment, rinse mouth with water and contact local poison control immediately (CarlRoth 2015).

Ethical issues: none

Environmental issues:

Lambda-cyhalothrin can cause serious harms to all organisms exposed to it. Prepare only the required volume for the experiment. As most municipal water treatment systems are not equip to remove lambda-cyhalothrin, collect all waste materials containing the pesticide into a glass container, do not pour anything down the sink, toilet, sewer, or street drain. After the experiment, get rid of the waste according to measures put up by local authorities (United State Environmental Protection Agency 2021).

Same calculation:

$$SEM = \frac{STDV}{\sqrt{number\ of\ trials}} \qquad STDV = \sqrt{\frac{\sum_1^N (x_i - \mu)^2}{N}}$$

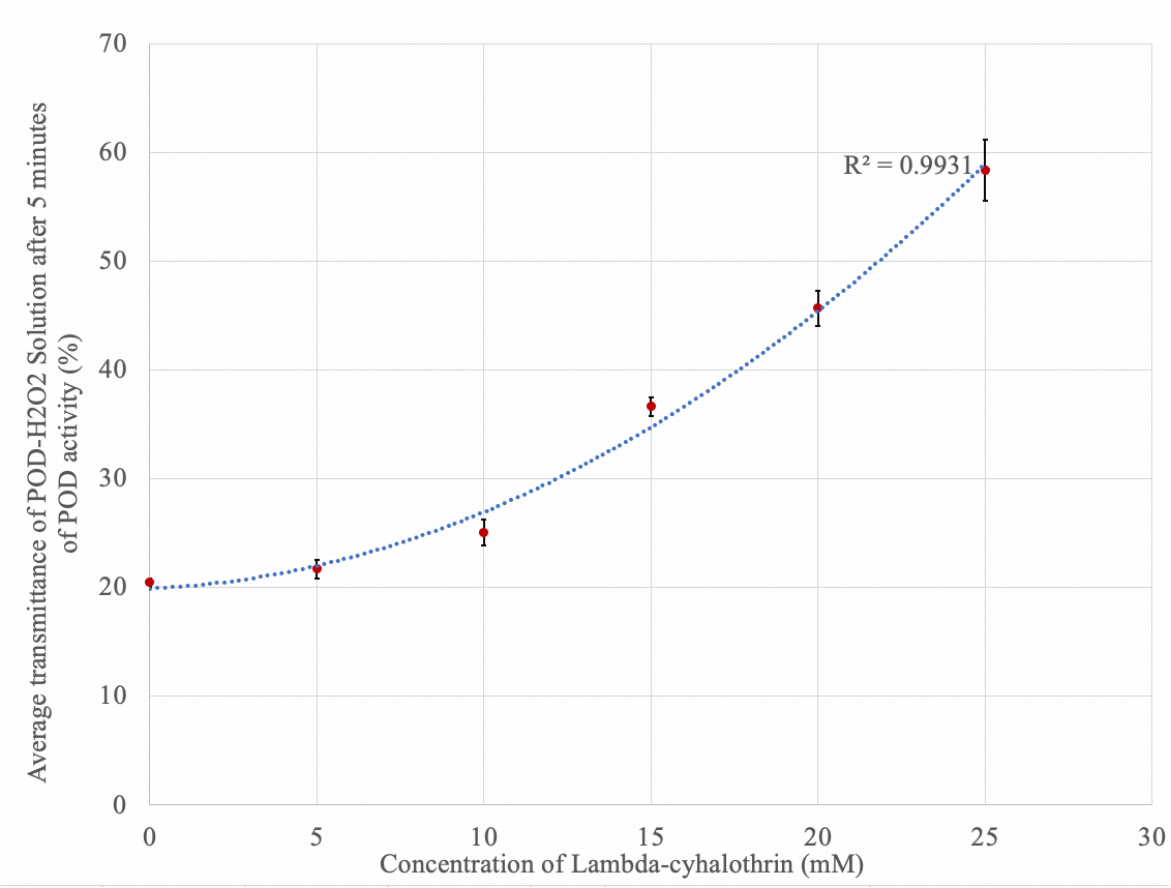
Raw data: Figure 1. Raw data table of the extent in which concentration of Lambda-cyhalothrin (mM) affects transmittance (%) of POD-H2O2 solution after 5 minutes of POD activities

Concentration of Lambda-cyhalothrin (mM)	Transmittance of POD-H2O2 Solution after 5 minutes of POD activity (%)				
	Trials				
	1	2	3	4	5
0 (control)	18. 89	20.56	19.28	21.36	20.64
5	20.33	21.78	20.32	20.54	22.44
10	27.12	28.33	25.89	26.94	25.91
15	36.58	35.32	38.11	36.29	37.84
20	41.99	43.21	47.47	42.78	44.96
25	47.5	50.41	56.32	53.19	49.43

Processed data: Figure 2. Processed data table of the extent in which concentration of Lambda-cyhalothrin (mM) affects transmittance (%) of POD-H2O2 solution after 5 minutes of POD activity

Concentration of Lambda-cyhalothrin (mM)	Average transmittance of POD-H2O2 Solution after 5 minutes of POD activity (%)	Standard deviation	Standard Error of the Mean
0	20.46	0.87	0.39
5	21.68	1.90	0.85
10	25.04	2.62	1.17
15	36.63	1.85	0.83
20	45.68	3.67	1.64
25	58.37	6.22	2.78

Figure 3. Graphical representation of the extent in which concentration of Lambda-Cyhalothrin (mM) affects average transmittance (%) of POD-H2O2 solution after 5 Minutes of enzyme activities *error bar for 0mM is there but too small to be visible*



Measurement of uncertainty:

Material Measured	Inaccuracy in Measurement	Impact on data
Radish POD	Measured with a graduated cylinder with an inaccuracy of ±0.1 ml (Indigo Instrument).	Affects POD concentration which leads to either an increase/decrease in oxygen production rate. This affects transmittance.
Lambda-cyhalothrin (or distilled water for control group)	Measured with a graduated cylinder with an inaccuracy of ±0.1 ml.	Affects concentration of Lambda-cyhalothrin which would lead to either more or less POD inhibition. This affects transmittance.
pH 7 buffer	Measured with a graduated cylinder with an inaccuracy of ±0.1 ml.	Affects pH of mixture causing it to deviate from the optimal pH of 6. This decreases POD actives leading to a lower than expected transmittance.
Guaiacol	Measured with a syringe with an inaccuracy of ±0.001ml (Thorbani & Stewart 1992).	Affects concentration leading to more or less reaction with oxygen. Changes in the production of GDHP affects transmittance.
Hydrogen peroxide	Measured with a syringe with an inaccuracy of ±0.001ml	Affects concentration which changes the moles of oxygen produced. This impacts transmittance.
	Vernier spectrophotometer inaccuracy not publicized	Affects data for transmittance

Interpretation of data:

Trend of data: According to Figure 3 as Lambda-cyhalothrin concentration increases from 0 mM to 10 mM, average transmittance increases slightly. From 10mM to 25mM there's a more drastic increase in transmittance. There is an overall strong positive relationship between Lambda-cyhalothrin concentration and transmittance. The R^2 value is 0.99 meaning only about 1% of the data does not fit this line. Data from Figure 2 shows that increase in average transmittance from 0mM of Lambda-cyhalothrin to 10 mM is low (only 1.22% difference). In comparison, the increase in transmittance is highest from 20mM to 25mM. The difference in transmittance between 0 mM of Lambda-cyhalothrin and 25 mM of Lambda-cyhalothrin is 37.91.

Variation in data: According to Figure 2, the standard deviations of experimental groups 0mM, 5mM, and 15 mM are relatively low (below 2% deviation) meaning that for these groups, each trial result varied at most by 2% from the average data making the mean a good presentation for the results of each trial. In comparison, the standard deviations of experimental groups 10mM, 20mM, and 25mM are much higher. This indicates a higher degree of variation from the average, making it a less accurate representation of the results of the trials. Overall, high standard deviation for 3 out of 6 experimental groups weakens the relationship between IV and DV. Out of the experimental groups, 0mM of Lambda-cyhalothrin has the least variation (0.87 %deviation) in data whereas data from the group with 25mM Lambda-cyhalothrin is the most varied (6.22% deviation). The fact that the group with 0 mM of Lambda-cyhalothrin has the least varied data indicates that it is the presence of Lambda-cyhalothrin which causes data variation. This is further supported by the fact that there is the most variation in data in the group with the highest concentration of Lambda-cyhalothrin (25 mM). Combining the two observations, Lambda-cyhalothrin's effects on transmittance appears to be inconsistent, especially when Lambda-cyhalothrin concentration is high. This inconsistency in transmittance rate when Lambda-cyhalothrin concentration is high can also be found when looking Figure 1. The difference between the lowest measured transmittance and the highest measured transmittance appears to grow as concentration increase. Indicating that as concentration of Lambda-cyhalothrin increases, the effect it has on transmittance becomes less and less predicable.

When looking at the error bars from Figure 3, it is clear that they are larger when concentration of Lambda-cyhalothrin is higher and smaller when concentration of Lambda-cyhalothrin is lower. Error bar represents an uncertainty in the means, therefore the previously mentioned growing unpredictability of transmittance as as concentration of Lambda-cyhalothrin increases persist. There is also an overlap data between 0mM and 5mM of Lambda-cyhalothrin. This demonstrates that the differences between the average transmittance of the two groups are not statistically significance, implying that when concentration of Lambda-cyhalothrin is below and at 5mM, it's effect on transmittance is negligible.

The first null hypothesis is that there this is no significant difference between experimental group: 0mM and 5mM. An unpaired T-test was performed between the control group (0mM) and 5mM Lambda-cyhalothrin. The p value was 0.3505. This indicates that the difference between the data set is not statistically significant, because the p value is above 0.05. The null hypothesis is proven to be correct.

The second null hypothesis is that there is no significant difference between experimental groups: 0mM and 25mM. An unpaired T-test was performed between the two groups. The p value was less than 0.0001. This indicates that the difference between the data set is extremely statistically significant because the p value is below 0.05. The null hypothesis is proven to be wrong.

Conclusion:

According to the data collected, the hypothesis that as lambda-cyhalothrin concentration increases, and the transmittance of the peroxide-peroxidase solution also increases after 5 minutes of reaction between POD and hydrogen peroxide is supported. This is made apparent when looking at Figure 3. As previously mentioned, a strong positive relationship (99% of the data points aligned with the trend line) is discovered between the IV and DV. However, at 5mM, the effects of lambda-cyhalothrin are not statistically insignificant with a 35% chance than the difference between 0mM and 5mM is due to chance (refer to first T-test). Similarly, transmittance at 10mM is also low. In comparison, from 10 to 25 mM the impact of lambda-cyhalothrin on transmittance is much more observable. Between 0mM and 25mM groups, there is less than 1 % chance that the difference between the data is due to chance (refer to second T-test). The reason for the observed positive relationship between lambda-cyhalothrin concentration and transmittance can be explained by lambda-cyhalothrin's inhibitory effect on POD. This is due the presence of a CN group in it chemical structure (Gulcin *et al*. 2018). Cyanide works by replacing the water in the six octahedral position of POD's iron heme group. The previously cited research by Chen *et al*. demonstrates irreversible inhibition of horseradish POD actives as a result of this process (2000). Lambda-cyhalothrin CN group's interactions with peroxidase's active site is an example of a chemical reaction. According to collision theory, a chemical reaction occurs when two reactants effectively collide with each other. The success of such process is dependent on chance. To increase collision rate one can increase reactant concentration, this increases the numbers of collisions between reactants (Key 2005). The same theory can be applied to Lambda-cyhalothrin cyanide's interaction with POD's iron heme group. As lambda-cyhalothrin concentration increases, the chances of collision between its CN group and POD active site increases. Thus, the inhibition of POD by the pesticide occurs faster. This means after five minutes of interaction, more enzymes will be inhibited by the pesticide. Once inhibited, POD can no longer catalyze H_2O_2 decomposition into oxygen and water. As transmittance rate is a measurement of oxygen formation (lower when more oxygen is formed), when lambda-cyhalothrin concentration increases and oxygen formation slows down, transmittance rate increases. The fact that 5mM of lambda-cyhalothrin in a solution has no observable effect on transmittance indicates that the collisions between the pesticide and POD active sites rarely occur and its inhibition effect is negligible. Only above 10mM lambda-cyhalothrin concentration does the effects of the pesticide on POD becomes apparent, indicating that from this point

on lambda-cyhalothrin has significant inhibition effect on POD. At 25mM the chances that lambda-cyhalothrin will inhibit POD activities is incredibly high, pointing to high numbers of collisions.

The fact that as lambda-cyhalothrin concentration increases, its impact on transmittance becomes more varied can be the result of uneven dissolution. Research conducted by Acevedo *et al*. indicates that as solute concentration increases fluctuation in regional solute concentration also increases (2018). Therefore, as lambda-cyhalothrin concentration increases, its distribution in the POD-H_2O_2 solution becomes less even. Thus in some cases, POD is exposed to higher than intended concentration of lambda-cyhalothrin while in others, POD is exposed to lower than intended concentration of lambda-cyhalothrin, resulting in varied degree of inhibition.

Strength and limitations:
One strength of the experiment is the DV is measured with a spectrophotometer. The machine provides precise numerical values which results in accurate data. Another strength of this experiment is that the methodology eliminates inconsistency in the volume of solution used. For liquids measured with the graduated cylinder, all measurements were taken by looking at the bottom of the meniscus. There were no air bubbles in the syringe and foam from Lambda-cyhalothrin is collected using a pipette. All these measures improve the accuracy of the experiment. Also, all test tubes and cuvettes were cleaned and dried before the solutions were poured in. This prevents contamination which would impact the results of the experiment.

Limitation	Its impact	Improvement for next time
Lambda-cyhalothrin is slightly opaque	As concentration of lambda-cyhalothrin increases, the pesticide becomes less and less transparent. This decreases the transmittance of the solution. This means transmittance becomes a slightly less accurate indicator for oxygen production as lambda-cyhalothrin concentration increases.	When measuring the transmittance of lambda-cyhalothrin at *X* mM, calibrate the spectrophotometer with a 2/3 full cuvette containing just lambda-cyhalothrin at *X* mM before recording data for the experimental group.
Uneven distribution of lambda-cyhalothrin	As lambda-cyhalothrin concentration increases, its distribution becomes less even. Therefore, at higher concentrations, data about lambda-cyhalothrin's impact on POD is less accurate.	After combining the two test tubes (refers to step 4 of Procedure for Lab), stir the mixture thoroughly with a glass mixer.

Extension:
A potential extension would be an investigation on how increasing concentration of Lambda-cyhalothrin affects a different root vegetable POD (ie. potatoes). The IV values should be kept constant for this experiment. Such an investigation can help provide deeper understanding into the similarities and differences between different types of root vegetable PODs. It can also demonstrate how dangerous Lambda-cyhalothrin is for root vegetables.

Another possible extension would be a study into how high concentration of Lambda-cyhalothrin needs to be fully stop radish POD activities. The IV value will be increased (25mM, 30mM, 35mM, 40mM, and 45mM) to see if Lambda-cyhalothrin at these concentrations will result in a transmittance of 100%.

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